

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness for individual datapoints) carries moderate validity risk because MILU does not document annotator demographics or inter-annotator agreement. Annotation relied primarily on subject experts at online exam portals [Q26] with manual audit by AI4Bharat volunteers [Q87]; institutional base is IIT Madras + IBM Research India [Q10], which does not guarantee Hindi-medium North Indian educational representation. The regional YAML notes 'no published MILU source provides annotator demographic details or inter-annotator agreement statistics'. The elicitation does mitigate this concern: the user confirmed evaluation schemas are 'broadly consistent across boards and competitive-exam settings' [A4], and dataset analysis shows cross-language answer-key consistency for parallel items [DATASET-D18, D19]. However, with ~all sampled Hindi validation items machine-translated [DATASET-D28], the original answer keys derive from English-source items and were applied to GPT-4O translations without documented Hindi-medium subject-expert audit of the translation-key alignment.

ID	Evidence
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
Q10	"Nilekani Centre at AI4Bharat, Indian Institute of Technology, Madras, IBM Research, India" (p.1)
D18	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1
D28	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... is_translated: True — is_translated=True flag on all 26 Hindi validation examples reviewed
D19	1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check

Strengths

- Subject experts at online exam portals provided initial answer accuracy [Q26]
- AI4Bharat volunteer team performed manual audits [Q87]
- Cross-language answer-key consistency observed for parallel items [DATASET-D18, D19]
- Elicitation confirms low expected divergence between competitive-exam keys and North Indian board norms [A4]

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Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels derive from exam answer keys [Q12, Q26]; the elicitation confirms these are broadly compatible with North Indian teacher norms [A4]. However, no Hindi-medium North Indian stakeholder validation is documented.

OC-2: Disagreement risk is moderate: most items (factual GK, science, history) are unlikely to have culturally variable answers, but translated items where GPT-4O may have introduced subtle translation errors [DATASET-D14, D15, D23] could yield Hindi-translation answer-key errors not caught by audit.

OC-3: INSUFFICIENT DOCUMENTATION — MILU does not provide annotator demographic breakdowns (regional background, Hindi-medium vs. English-medium education, subject expertise) [NOT FOUND per regional YAML annotator_demographics]; no inter-annotator agreement statistics are reported.

OC-4: Re-annotation by Hindi-medium North Indian teacher pool would address the documented gap, particularly for translated items and Arts & Humanities content where register matters most.

OC-5: INSUFFICIENT DOCUMENTATION — aggregation methods for the cluster-to-subject manual merging [Q41] do not report how disagreements among annotators were resolved, leaving minority-perspective handling opaque.

OC-6: Documented label issues: (a) annotator demographics opaque; (b) no inter-annotator agreement; (c) translated items inherit English-source keys without documented Hindi-medium subject-expert validation.

ID	Evidence
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D15	रक्त में नमिलनखिति में से कसिकी अधकिता 'ब्लू बेबी सड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D23	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
WEB-10	NAACL 2025 paper does not publish annotator demographic details or inter-annotator agreement (https://aclanthology.org/2025.naacl-long.507/)

Information Gaps

- Annotator demographics (region, language background, subject expertise)

- Inter-annotator agreement statistics
- Whether translated Hindi items underwent Hindi-medium subject-expert key validation
- Disagreement-resolution protocols in cluster-to-subject manual merging

Requires Expert Verification

- Spot-check a sample of translated Hindi answer keys against North Indian teacher judgment
 - Verify that competitive-exam keys for Hindi literature, civics, and regional history items align with state-board mark schemes
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Remediation

Gap 1: No annotator demographics or inter-annotator agreement published; translated Hindi items inherit English-source keys without documented Hindi-medium subject-expert audit

Recommendation: Conduct a focused re-annotation audit by Hindi-medium North Indian teachers on a stratified sample (~200 items) of MILU Hindi items, prioritising Arts & Humanities, Law & Governance, and translated science items to estimate label-correctness rates